

Sarvan A

Blockchain Engineering – R&D

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PROFESSIONAL OVERVIEW

Blockchain Engineer with 2.9+ years building and securing DeFi protocols across EVM chains and Solana. Core strengths include Solidity and Rust (Anchor) smart contract development, AMM logic, and gas optimization — backed by Slither security audits and property-based test coverage above 90%. Comfortable owning a feature end-to-end, from on-chain program to the React/Web3 frontend that talks to it.

Technical Skills

- **Smart Contract Development:** Solidity (v0.8.x), Rust, EVM, Solana, Go (Golang), JavaScript, TypeScript
- **Blockchain & Runtimes:** DeFi Protocols, AMM Design, Liquidity Pool Architecture, Foundry, Hardhat, Anchor Framework, OpenZeppelin Contracts, Solana CLI, Remix IDE
- **Web3 Client Integration:** Ethers.js, Web3.js, @solana/web3.js
- **Security & Testing:** Security Audits, Slither, Aderyn, Unit & Invariant Testing, Property-Based Fuzzing, Gas Optimization & Profiling (hardhat-gas-reporter)
- **Backend & Systems:** Node.js, Express, Gin-Gonic, PostgreSQL, Redis, MongoDB, Docker, Git, CI/CD

WORK EXPERIENCE

Osiz Technologies Private Limited - Madurai, India

BLOCKCHAIN ENGINEER (R&D)

JAN 2024 – PRESENT

- Architected, tested, and deployed secure EVM smart contracts and Solana on-chain programs for decentralized finance and Web3 applications.
- Engineered custom **DeFi** Automated Market Maker (AMM) protocols adapted from Uniswap and PancakeSwap V2 architectures, implementing custom swap routing, dynamic LP fee distribution, and non-reentrant liquidity staking pools.
- Implemented automated testing workflows with Foundry and Hardhat, elevating branch test coverage to over 90% across production contract deployments.
- Conducted **security audits** using Slither and Aderyn, identifying and eliminating vulnerability classes, including reentrancy vectors and access-control flaws.
- Optimized Solidity storage layouts through variable packing and custom error architectures, decreasing runtime gas consumption by 15% to 22% on high-frequency transactions.
- Integrated smart contract interfaces with React.js client frontends via Ethers.js and Web3.js, standardizing multi-provider wallet connections across MetaMask, Phantom and WalletConnect.

PROJECTS

Redemption 1987 - Solana On-Chain Vault & Token Distribution Program

- Developed an on-chain token vault program in Rust using Anchor to execute trustless asset redemptions and scheduled distributions.
- Integrated program instructions with a React.js dApp via @solana/web3.js, implementing compute budget optimizations to prevent simulation and execution drops.
- Executed Cross-Program Invocations (**CPI**) targeting the SPL Token Program for programmatic token transfers, minting, and burn mechanisms.

Bitroleum Ecosystem

- Designed and deployed a **DeFi** AMM protocol powering token swaps, liquidity pools, and staking reward emissions.
- Secured routing and reward contracts against sandwich and reentrancy attacks using OpenZeppelin ReentrancyGuard and two-step ownership patterns.
- Engineered a high-concurrency microservice in Go (Gin-Gonic) using go-ethereum (ethclient) to index on-chain swap and sync events, streaming real-time reserve balances, volume, and dynamic APY metrics to Redis.
- Built a responsive trading interface in React.js featuring real-time slippage management, deadline checks, and balance polling.

Peacock NFT

- Architected an **ERC-1155** multi-token contract supporting batch minting, batched transfers, and semi-fungible lifecycle management.
- Developed a Go (Gin-Gonic) backend pipeline to automate off-chain IPFS metadata pinning and cryptographic signature generation (**EIP-712**) for gasless whitelisted minting verification.
- Connected contract state to a React.js client dApp via Web3.js, incorporating automated event listeners and transaction receipt handling.
- Integrated decentralized asset metadata storage with IPFS, ensuring immutable on-chain token URI resolution.

EDUCATION

Master of Computer Applications (MCA)

Arul Anandar College, Madurai, India | 2021 – 2023 | Aggregate: 84.5%

Bachelor of Science in Computer Science (B.Sc. CS)

Arul Anandar College, Madurai, India | 2018 – 2021 | Aggregate: 64.8%